

EchoVoice

An On-Device Speech Recognition and Pronunciation Assessment App for Listen-and-Repeat Speech Therapy

English Data Dictionary (Level 0, Level 1, and Level 2 DFDs)

External Entities

People outside the system who send data in or receive data out.

Entity	Plain-English Definition	Sends to System	Receives from System
Learner	The individual (child or adult) undergoing speech therapy who listens to a prompt and repeats it into the device microphone.	Speech Input	Visual & Auditory Prompts
Caregiver	A parent or guardian who chooses which exercise the Learner practices and receives a plain-language summary of how the session went.	Exercise Selection	Session Progress Summary
SLP (Speech-Language Pathologist)	The clinician who sets therapy goals and phoneme priorities and reviews the app's automated assessments.	Goal Configuration & Assessment Oversight	Longitudinal Progress Reports

Data Stores

Places where the system keeps data between processes.

ID	Name	Plain-English Definition
D1	Progress Database	Holds every completed assessment record (score, phoneme errors, timestamp) for a Learner, used to compute trends over time.
D2	Goal Config Store	Holds the therapy goals and phoneme priorities the SLP has configured — what the system currently treats as "active goals."
D3	Phoneme Target Dictionary	Reference table of the correct/expected phoneme pattern for each target word, used as the answer key during scoring.

Data Flows

Every labeled arrow on the diagrams, in plain English, listed alphabetically.

Data Flow	Plain-English Definition
Active Goals	The current set of phonemes/exercise priorities pulled from D2, used to pick which words the Learner practices.
Assessment Record	The saved result of one attempt — learner, exercise, timestamp, phoneme errors, and score — written to D1.
Assessment Result	The scored outcome of a single attempt, passed on so feedback can be generated for the Learner.
Exercise Selection	The Caregiver's choice of which exercise or word list the Learner will practice.
Goal Configuration & Assessment Oversight	The SLP's clinical input: setting/updating therapy goals and checking over the app's assessments.

Data Flow	Plain-English Definition
Longitudinal Progress Reports / Progress Report	A trend view of the Learner's accuracy over time, sent to the SLP for clinical review.
Phoneme Error Matrix	A phoneme-by-phoneme comparison showing which sounds were correct, substituted, omitted, or distorted.
Predicted Phoneme Sequence	The sequence of phonemes the on-device recognition model believes the Learner said.
Raw Audio File (.wav)	The unprocessed recording of the Learner's spoken attempt.
Recorded Speech	The captured audio handed from speech capture to the recognition/assessment process.
Session Progress Summary	A simple, non-clinical recap (e.g., accuracy, streaks) of the Learner's recent session, sent to the Caregiver.
Speech Input	The Learner's spoken repetition of the prompted word or phrase.
Spectrogram Tensors	The numeric, model-ready representation of the audio, produced from the raw recording.
Store Goal Configuration	The action of saving newly set or updated therapy goals into D2.
Stored Progress	Past assessment records pulled from D1 to build trends and summaries.
Target Phoneme Template	The correct/expected phoneme pattern for the current word, retrieved from D3 for comparison.
Target Word	The specific word or phrase chosen for the current practice attempt.
Visual & Auditory Prompts	The on-screen text/image and spoken audio of the target word shown to the Learner.

Processes

What each numbered bubble on the diagrams actually does.

ID	Name	Plain-English Definition
0	EchoVoice Application	The entire app, viewed as a single black box: takes speech and configuration in, gives prompts and progress reports out.
1.0	Manage Exercise	Picks the next target word based on the Caregiver's exercise choice and the SLP's active goals, then prompts the Learner.
2.0	Capture Speech	Plays the prompt and records the Learner's spoken attempt as an audio file.
3.0	Speech Recognition & Assessment	Runs the recording through recognition and scoring to produce an assessment (see 3.1-3.4 for the internal steps).
4.0	Generate Feedback	Turns the assessment result into a cue the Learner can understand (praise, encouragement, or a correction hint).
5.0	Monitor Progress	Looks at stored history to build a caregiver-friendly summary and a clinician-facing progress report.
6.0	Configure Goals & Review Assessment	Lets the SLP set therapy goals (saved to D2) and review assessment records; publishes the active goals to the rest of the app.
3.1	Extract Acoustic Features	Converts the raw audio recording into a spectrogram the recognition model can read.
3.2	Run Model Inference	Runs the on-device model on the spectrogram to predict which phonemes were spoken.
3.3	Align Phonemes	Compares the predicted phonemes to the correct target pattern and marks each one right, wrong, missing, or distorted.
3.4	Compute Pronunciation Score	Turns that comparison into a numeric score, sending it onward for feedback and saving it for the record.

